Cover Letter

Date: 2026-09-23 To: Editor-in-Chief, Advances in Space Research From: LIU Shuhao, College of Intelligent Systems Science and Engineering, Harbin Engineering University, shuhao.liu.86@outlook.com

Dear Editor,

I am submitting the manuscript "Physics-Aware Multi-Objective Scheduling for an Agile SAR Satellite: Balancing Profit, NESZ, and Geometric Resolution" for consideration as a full-length article in Advances in Space Research.

This manuscript falls within the journal's scope of space technology applications for Earth observation: it studies the scheduling of an agile SAR satellite whose imaging quality is coupled to observation geometry, a problem at the intersection of spacecraft resource management and SAR remote sensing instrumentation.

The manuscript presents a physics-aware multi-objective optimization framework for single-satellite agile SAR scheduling. Five configurations spanning greedy heuristics, a single-objective GA control, and NSGA-III variants are compared on 200 systematically varied scenarios across four density classes (N = 20 to 500). In sparse request sets, re-optimizing observation times buys measurable image quality at a stated coverage price: +40% radiometric-index quality at a 15% coverage cost and +33.7% geometric-index quality at 14%, with a matched-knee-point control confirming that the third objective contributes a modest controlled +4.9% margin rather than an artifact of the NSGA-III reference-direction count or knee selection. Two measured boundaries delimit that gain. It contracts from +28% to +2.3% in geometric quality as the platform's squint envelope narrows from 45 to 15 degrees, and it vanishes beyond roughly 150 targets, where G-BL-seeded NSGA-III converges to the greedy coverage anchor with quality margins below 2%, conditional on greedy hot-start initialization. The work establishes when multi-objective SAR quality scheduling is worth its coverage cost, within the single-satellite, single-pass Stripmap setting studied.

Disclosure of related manuscript: I have an article previously accepted by Advances in Space Research entitled "Event-Triggered Attitude Decision-Making for Agile SAR Satellite Formations: A Self-Triggered Replanning Approach" (Ch4 of the author's doctoral thesis). That work addresses event-triggered attitude control decisions for SAR satellite formations, which is an orthogonal topic to the present manuscript. The two works share no overlapping experimental results, solver implementations, or research claims. The current manuscript focuses on single-satellite scheduling with imaging quality objectives; the accepted work addresses formation-level attitude replanning triggered by observation events. I confirm that the present manuscript contains no text or figures from the accepted article, and that the two manuscripts are distinct contributions to the author's doctoral thesis. Elsevier's same-journal policy permits multiple articles from the same thesis when they address different aspects of the research.

This manuscript is original, has not been published elsewhere, and is not under

consideration by another journal. All authors have approved the submission.

Thank you for your consideration.

Sincerely,

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